

# Module\_1: Team ELMQVIST 7

## Team Members:

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## Project Title:

Evaluating Years Spent in Education as Mediator for Symptom Onset-age

## Project Goal:

Neuroscientists have found that learning can physically alter the brain. A study by Asano, Taki, Hashizume et al. (2014) conducted a longitudinal study of 229 healthy children in Japan between the ages of 5 and 18. They aimed to see whether the amount of time children studied at home was related to the amount of white and gray matter. They did an MRI scan of every participant at the beginning of the study and one at the end, 3 years later. They found a positive correlation between studying 1+ hour a night and increased white matter in the right precuneus and right superior temporal gyrus. Furthermore, they found a positive correlation with change in gray matter in the right precuneus and the right inferior parietal lobe (Asano, Taki, Hashizume et al. 2014). Other neuroscientists have found that the right superior frontal gyrus is known to play an important role in memory. Furthermore, increased white and gray matter is known to be a sign of an increased number of axons. Therefore, we hypothesize that if learning increases the number of axons in the areas of the brain that are important for memory, the memory loss symptom of Alzheimer's (dementia) will be delayed, and the age at onset will be larger. This article inspired us to investigate whether the number of years of education a patient had impacted the age at which their Alzheimer's symptoms started. This can be determined by comparing the years spent in education and age of onset for each patient. We are also curious if this relationship would change depending on the patient's genetic predisposition. We can investigate this by creating subgroups filtered by the patient's version of the apolipoprotein E (APOE) gene. The 3 variants of APOE are  $\epsilon 2$ ,  $\epsilon 3$ , and  $\epsilon 4$ . Every human has two of these variants, making 6 possible combinations: 2/2, 2/3, 2/4, 3/3, 3/4, and 4/4. Individuals with an  $\epsilon 4$  allele have a higher risk of getting AD, and patients with two  $\epsilon 4$  alleles (4/4) are at the highest risk.

## Research questions:

- 1) Does the length of time (years) an individual spends being educated impact the age at which their Alzheimer's symptoms begin?
- 2) Does the length of time (years) an individual spends being educated play a different role in onset symptoms depending on the individual's genetic predisposition?

## Disease Background:

### Prevalence and Incidence — United States

**Prevalence** is the number of people living with a disease at a given time.

**Incidence** is the number of new cases that occur over a specific period.

#### Prevalence

- **About 7.4 million** Americans age 65 and older are living with Alzheimer's disease (AD), according to the estimate in our notes.
- The number of Americans living with AD is projected to reach **13.8 million by 2060**.

#### Incidence

- Our notes report approximately **900,000 adults diagnosed with Alzheimer's annually**. Confirm the source, year, and population covered; incidence is best described as *new cases per year*, rather than an incidence rate unless a population denominator is provided.

#### Related mortality statistic

- Our notes report a **142% increase in deaths among people with AD** while deaths from other diseases declined. Add the comparison years and verify the statistic before including it as a finding.

**Sources:** [Alzheimer's Association — Facts and Figures](#); [PubMed Central article](#)

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## Economic Burden

Alzheimer's disease (AD) creates costs for families, caregivers, and the health care system.

- In **2025**, about **12 million** caregivers provided **19.2 billion hours** of unpaid care to people with dementia.
- Unpaid care for people with AD was valued at **\$413.5 billion in 2024**.
- Health care, long-term care, and hospice services for people with dementia cost an estimated **\$384 billion in 2025**.
- Caregiving can also affect caregivers' physical and mental health.

**Source:** [Alzheimer's Association — Facts and Figures](#)

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## Risk factors (genetic, lifestyle)

Factors associated with a higher risk of Alzheimer's disease include:

- **Physical health:** Physical inactivity, obesity, high blood pressure, and diabetes.
- **Lifestyle:** Smoking and excessive alcohol use.
- **Mental and sensory health:** Depression and hearing loss.
- **Genetics and demographics:** Family history and sex; women have higher rates of Alzheimer's.

These factors may increase risk, but they do not mean someone will develop Alzheimer's disease.

**Source:** [Alzheimer's Association — Causes and Risk Factors](#)

**AI use:** Gemini & Google AI helped find credible sources and polish this section. The VS Code agent helped format it.

## Societal determinants

Social and environmental conditions can affect brain health and access to dementia care. These include:

- **Socioeconomic factors:** Education, income, and access to health care.
- **Neighborhood factors:** Air pollution and access to healthy food.
- **Social factors:** Discrimination and other barriers to care.

**Sources:** [CDC](#) · [DC Brain Health](#) · [Research article](#)

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## Symptoms

Common symptoms of Alzheimer's disease include:

- **Memory loss** that disrupts daily life.
- **Difficulty planning or solving problems**, such as following familiar tasks or working with numbers.
- **Visual and spatial difficulties**, including trouble judging distance or knowing where you are.

**Source:** [CDC — Signs and Symptoms of Alzheimer's Disease](#)

**AI use:** Gemini & Google AI helped find credible sources and polish this section. The VS Code agent helped format it.

## Diagnosis

Alzheimer's disease is diagnosed through a medical evaluation. This may include:

- **Medical history and cognitive tests** to assess memory and thinking.
- **Physical and neurological exams** to check for other possible causes of symptoms.
- **Lab tests and brain scans**, such as MRI, CT, or PET scans.
- **Biomarker tests**, including tests of cerebrospinal fluid, when appropriate.

**Source:** [National Institute on Aging — How Is Alzheimer's Disease Diagnosed?](#)

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## Standard Treatments and Coverage

Treatment depends on the stage of Alzheimer's disease. Medicines may help manage symptoms or slow decline; they do not cure the disease.

| Treatment                                   | Use                                                                                                                                            | Common risks or side effects                                |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Donepezil, galantamine, rivastigmine</b> | Help manage symptoms; galantamine and rivastigmine are generally used for mild to moderate disease. Donepezil may be used at different stages. | Nausea, diarrhea, or loss of appetite                       |
| <b>Memantine</b>                            | Helps manage symptoms in moderate to severe disease.                                                                                           | Dizziness, headache, or constipation                        |
| <b>Lecanemab and donanemab</b>              | For some people with early Alzheimer's disease; may slow cognitive decline.                                                                    | Brain swelling or bleeding (ARIA); MRI monitoring is needed |
| <b>Brexpiprazole</b>                        | May treat agitation associated with Alzheimer's disease.                                                                                       | Drowsiness, dizziness, or restlessness                      |

**Coverage:** Insurance coverage and out-of-pocket costs vary by medication and plan. Check with the insurer and health care provider about eligibility and costs.

**Sources:** [National Institute on Aging — Alzheimer's Disease Treatment](#) · [CMS — Medicare Coverage of Alzheimer's Drugs](#)

**AI use:** Gemini & Google AI helped find credible sources and polish this section. The VS Code agent helped format it.

## Disease progression & prognosis

Alzheimer's disease progresses differently for each person. On average, people live about **8 years after symptoms begin**, though some live for **20 years or longer**. Early-onset Alzheimer's is rare and can begin as early as a person's 30s or 40s.

The stages of progression include:

1. **Preclinical stage:** Brain changes begin before noticeable symptoms appear.
2. **Early stage:** Mild memory and thinking problems may affect daily activities.
3. **Middle stage:** Symptoms worsen, and the person may need more help with daily activities.
4. **Late stage:** Severe changes in memory and physical abilities may require full-time care.

**Source:** [Johns Hopkins Medicine — Stages of Alzheimer's Disease](#)

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## Continuum of Care Providers

People with Alzheimer's may need different types of support as their needs change. Care options can include:

- **In-home services:** Help with daily activities and personal care while living at home.
- **Assisted living:** Housing with support for daily activities, meals, and medication routines.
- **Memory care:** Specialized support in a setting designed for people with memory conditions.
- **Long-term care:** Ongoing assistance for people who need substantial help. It may be provided at home or in a residential facility.

Care needs vary, so people may move between services or use more than one type of care.

**Sources:** [Understanding the Continuum of Care](#) · [LCCA — Continuum of Care](#)

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## Biological mechanisms (anatomy, organ physiology, cell & molecular physiology)

Alzheimer's disease causes changes in the brain that damage neurons and disrupt their connections. Two key changes are:

- **Amyloid plaques:** Amyloid-beta proteins build up between neurons.
- **Tau tangles:** Tau proteins accumulate inside neurons, disrupting their function.

As the damage spreads, neurons lose connections and die. This can lead to shrinkage in affected brain areas and problems with memory, thinking, and other abilities. The exact sequence of changes is still being studied.

**Sources:** [National Institute on Aging — What Happens to the Brain in Alzheimer's Disease?](#) · [YouTube video](#)

**AI Use:** Gemini & Google AI helped find credible sources and polish this section. The VS Code agent helped format it.

## Clinical Trials/next-gen therapies

Clinical trials evaluate potential Alzheimer's treatments for safety and effectiveness. Newer therapies aim to do more than ease symptoms; some target processes involved in disease progression. Researchers continue to study who may benefit, how well treatments work, and what risks they carry. A treatment discussed in an article is not necessarily approved or proven effective—check its current trial status and consult reliable medical sources.

**Sources:** [Scientific American — A New Generation of Alzheimer's Treatments, Explained in Graphics](#) · [Cromos Pharma — Is Clinical Research Ready for the Next Era of Alzheimer's Trials?](#)

**AI Use:** Gemini & Google AI helped find credible sources and polish this section. The VS Code agent helped format it.

## Data-Set:

We used the class-provided **Metadata and Protein Data** file, which has 84 donors and 50 variables covering demographics, clinical information, neuropathology, and protein measurements taken from brain tissue after death. The data come from the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) and match the cohort in Gabitto et al. (2024), "Integrated multimodal cell atlas of Alzheimer's disease," *Nature Neuroscience*, 27, 2366-2383 (<https://doi.org/10.1038/s41593-024-01774-5>).

The donors were 84 older adults (51 female, 33 male, ages 65 to 102 at death) from two research studies, the Adult Changes in Thought (ACT) study and the University of Washington Alzheimer's Disease Research Center. They were chosen to cover the full range of Alzheimer's pathology. Brains were collected soon after death, and neuropathology was scored using standard staging methods and tissue staining. Details like APOE genotype, education, and symptom onset came from participant records and study visits.

For our question we compared **APOE genotype** (categories 2\_3, 3\_3, 3\_4, and 4\_4), **years of education** (in years), and **age of onset of cognitive symptoms** (in years). We wanted to see whether more education was linked to later symptom onset, and whether the APOE groups made a difference. We left out genotypes 2\_4 and 2\_2 because they only had 2 and 1 donors. Age of onset was only recorded for 38 of the 84 donors, so our comparison uses just those donors.

## Data Analysis:

### 1. Import models to calculate statistical test and generate graphs

```
#importing the constructors and class made in Mod1_patient
from Mod1_patient import Patient

#import the proper packages for creating the plots and standard deviation
import matplotlib.pyplot as plt
from scipy import stats
import numpy as np
import statistics
import pandas as pd
```

```
from sklearn.linear_model import LinearRegression
Patient.all_patients = [] #this line resets the list of all patients
so there is no accumulation of data from previous runs
```

2. Create list that will hold the values needed to compare the age of onset symptoms to years of education for each specific gene version

```
#These list will be used to create the scatter plots and bar graph
age_of_onset_4_4 = []
age_of_onset_3_4 = []
age_of_onset_2_4 = []
age_of_onset_3_3 = []
age_of_onset_3_2 = []
years_of_education_4_4 = []
years_of_education_3_4 = []
years_of_education_2_4 = []
years_of_education_3_3 = []
years_of_education_3_2 = []
```

3. Call method from Mod1\_patient.py that will read through the imported CSV file and creates objects according to the constructor in class Patient

```
#this calls the method created in the patient.py file that opens the
CSV file and goes through the data according to the following methods
filename = "/Users/janewyatt/Desktop/BME 2315 /Module
1/BME2315_Module1_Project/Metadata and Protein Data for Module 1.csv"
Patient.instantiate_from_csv(filename)

#display a sample of the data file with rows and columns
# df = pd.read_csv(
#     "/Users/janewyatt/Desktop/BME 2315 /Module
1/BME2315_Module1_Project/Metadata and Protein Data for Module 1.csv"
# )
# df.head()
```

4. Call the methods from Mod1\_patient.py that sort through the patients in the data file and returns only the ones with the requested gene combination. Adds these patient data points to the previously created list.

```
#Calling the method get_patients_with_4_4_gene to sort the patients in
the CSV file into only patients with 4_4 gene
print("\nList of All Patients with 4_4 Gene sorted by age of onset:\n")
for patient in Patient.get_patients_with_4_4_gene("4_4"):
    if patient.age_of_onset is not None and patient.years_of_education
```

```

is not None:
    #this condition checks to make sure the patient has both age
of onset and years of education data
    age_of_onset_4_4.append(patient.age_of_onset)
    years_of_education_4_4.append(patient.years_of_education)
    #these append functions add the values that meet the condition
into the proper list
    print(patient)
    #Printing the patients with 4_4 gene to keep the data organize

#This block of the code does the same thing as the one from lines 30-
39 but for the 3_4 gene
print ("\nList of All Patients with 3_4 Gene sorted by age of onset:\n")
for patient in Patient.get_patients_with_3_4_gene("3_4"):
    if patient.age_of_onset is not None and patient.years_of_education
is not None:
        age_of_onset_3_4.append(patient.age_of_onset)
        years_of_education_3_4.append(patient.years_of_education)
    print(patient)

#This block of the code does the same thing as the one from lines 30-
39 but for the 2_4 gene
print ("\nList of All Patients with 2_4 Gene sorted by age of onset:\n")
for patient in Patient.get_patients_with_2_4_gene("2_4"):
    if patient.age_of_onset is not None and patient.years_of_education
is not None:
        age_of_onset_2_4.append(patient.age_of_onset)
        years_of_education_2_4.append(patient.years_of_education)
    print(patient)

#This block of the code does the same thing as the one from lines 30-
39 but for the 3_3 gene
print ("\nList of All Patients with 3_3 Gene sorted by age of onset:\n")
for patient in Patient.get_patients_with_3_3_gene("3_3"):
    if patient.age_of_onset is not None and patient.years_of_education
is not None:
        age_of_onset_3_3.append(patient.age_of_onset)
        years_of_education_3_3.append(patient.years_of_education)
    print(patient)

#This block of the code does the same thing as the one from lines 30-
39 but for the 3_2 gene
print ("\nList of All Patients with 3_2 Gene sorted by age of onset:\n")
for patient in Patient.get_patients_with_3_2_gene("3_2"):
    if patient.age_of_onset is not None and patient.years_of_education
is not None:

```

```

        age_of_onset_3_2.append(patient.age_of_onset)
        years_of_education_3_2.append(patient.years_of_education)
    print(patient)

if len(years_of_education_3_2) < 2:
    print("Unfortunately there is not enough data for a 3_2
    regression.")
    #While writing the code we realized that there aren't any patients
    in the data set with a 3_2 gene and a recorded length of years spent
    in school or onset age

```

List of All Patients with 4\_4 Gene sorted by age of onset:

```

Gender: Female, Age of Onset: 88, Gene Type: 4_4
Gender: Male, Age of Onset: 64, Gene Type: 4_4
Gender: Female, Age of Onset: 64, Gene Type: 4_4
Gender: Female, Age of Onset: 63, Gene Type: 4_4
Gender: Female, Age of Onset: 53, Gene Type: 4_4
Gender: Male, Age of Onset: None, Gene Type: 4_4

```

List of All Patients with 3\_4 Gene sorted by age of onset:

```

Gender: Female, Age of Onset: 87, Gene Type: 3_4
Gender: Female, Age of Onset: 82, Gene Type: 3_4
Gender: Male, Age of Onset: 80, Gene Type: 3_4
Gender: Female, Age of Onset: 78, Gene Type: 3_4
Gender: Male, Age of Onset: 72, Gene Type: 3_4
Gender: Female, Age of Onset: 71, Gene Type: 3_4
Gender: Female, Age of Onset: 61, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Male, Age of Onset: None, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Male, Age of Onset: None, Gene Type: 3_4
Gender: Male, Age of Onset: None, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Female, Age of Onset: None, Gene Type: 3_4
Gender: Male, Age of Onset: None, Gene Type: 3_4

```

List of All Patients with 2\_4 Gene sorted by age of onset:

```

Gender: Female, Age of Onset: 88, Gene Type: 2_4
Gender: Male, Age of Onset: 82, Gene Type: 2_4

```

List of All Patients with 3\_3 Gene sorted by age of onset:

```

Gender: Male, Age of Onset: 94, Gene Type: 3_3
Gender: Male, Age of Onset: 93, Gene Type: 3_3

```



5. Create the scatter plots using the previously created list and converting them into arrays. Apply a linear regression analysis using the imported model.

```
#-----  
-----  
#Creating the scatter plots using the the list in lines 15-24  
  
#This is the scatter plot of 4_4 gene  
#At AI's recommendation we converted the data list into numpy arrays  
because the array assist in reformatting the data to make it easier to  
plot with the scikit-learn (the model that is performing the linear  
regression)  
X = np.array(years_of_education_4_4).reshape(-1, 1)  
y = np.array(age_of_onset_4_4)  
  
#this tells python wich data sets to use for the x and y values for  
the line in the graph  
model = LinearRegression()  
model.fit(X, y)  
  
#these statistics model caculate the slope, intercept, and R-squared  
value  
slope = model.coef_[0]  
intercept = model.intercept_  
r2 = model.score(X, y)  
  
equation = f"y = {slope:.2f}x + {intercept:.2f}\nR² = {r2:.2f}"  
  
#the following block of code displays the graph and the regression  
equation  
plt.scatter(X[:, 0], y, color="blue")  
plt.plot(X[:, 0], model.predict(X), color="red")  
plt.text(  
    #this formatting was generated by AI to keep the equation from  
being cut off when the graph was generated  
    0.05,  
    0.95,  
    equation,  
    transform=plt.gca().transAxes,  
    color="red",  
    verticalalignment="top",  
    bbox={"facecolor": "white", "alpha": 0.8, "edgecolor": "none"},  
)  
plt.xlabel("Years of Education")  
plt.ylabel("Age of Onset")  
plt.title("4_4: Years of Education vs Age of Onset")  
plt.show()  
  
#-----
```

*# Scatter plot for data points with gene 3\_4*  
*#the following code is the same as for lines 80-112 but for the 3\_4 gene*

```
X = np.array(years_of_education_3_4).reshape(-1, 1)
y = np.array(age_of_onset_3_4)
```

```
model = LinearRegression()
model.fit(X, y)
```

```
slope = model.coef_[0]
intercept = model.intercept_
r2 = model.score(X, y)
```

```
equation = f"y = {slope:.2f}x + {intercept:.2f}\nR2 = {r2:.2f}"
```

```
plt.scatter(X[:, 0], y, color="blue")
plt.plot(X[:, 0], model.predict(X), color="red")
plt.text(
    0.05,
    0.95,
    equation,
    transform=plt.gca().transAxes,
    color="red",
    verticalalignment="top",
    bbox={"facecolor": "white", "alpha": 0.8, "edgecolor": "none"},
)
plt.xlabel("Years of Education")
plt.ylabel("Age of Onset")
plt.title("3_4: Years of Education vs Age of Onset")
plt.show()
```

*#-----*

*# Scatter plot for data points with gene 2\_4*  
*#the following code is the same as for lines 80-112 but for the 2\_4 gene*

```
X = np.array(years_of_education_2_4).reshape(-1, 1)
y = np.array(age_of_onset_2_4)
```

```
model = LinearRegression()
model.fit(X, y)
```

```
slope = model.coef_[0]
intercept = model.intercept_
r2 = model.score(X, y)
```

```
equation = f"y = {slope:.2f}x + {intercept:.2f}\nR2 = {r2:.2f}"
```

```
plt.scatter(X[:, 0], y, color="blue")
plt.plot(X[:, 0], model.predict(X), color="red")
```

```

plt.text(
    0.05,
    0.95,
    equation,
    transform=plt.gca().transAxes,
    color="red",
    verticalalignment="top",
    bbox={"facecolor": "white", "alpha": 0.8, "edgecolor": "none"},
)
plt.xlabel("Years of Education")
plt.ylabel("Age of Onset")
plt.title("2_4: Years of Education vs Age of Onset")
plt.show()
#-----
# Scatter plot for data points with gene 3_3
#the following code is the same as for lines 80-112 but for the 3_3
gene

X = np.array(years_of_education_3_3).reshape(-1, 1)
y = np.array(age_of_onset_3_3)

model = LinearRegression()
model.fit(X, y)

slope = model.coef_[0]
intercept = model.intercept_
r2 = model.score(X, y)

equation = f"y = {slope:.2f}x + {intercept:.2f}\nR2 = {r2:.2f}"

plt.scatter(X[:, 0], y, color="blue")
plt.plot(X[:, 0], model.predict(X), color="red")
plt.text(
    0.05,
    0.95,
    equation,
    transform=plt.gca().transAxes,
    color="red",
    verticalalignment="top",
    bbox={"facecolor": "white", "alpha": 0.8, "edgecolor": "none"},
)
plt.xlabel("Years of Education")
plt.ylabel("Age of Onset")
plt.title("3_3: Years of Education vs Age of Onset")
plt.show()
#-----
# Scatter plot for data points with gene 3_2
# the following code is the same as for lines 80-112 but for the 3_2
gene

```

```

X = np.array(years_of_education_3_2).reshape(-1, 1)
y = np.array(age_of_onset_3_2)

if len(X) >= 2: #this conditional checks if there are at least 2 data
points in the group because if its not 2 points or more the linear
regression method will not work
    #because the 3_2 gene has less then 2 data points the regression
will not run and the graph will not be displayed
    model = LinearRegression()
    model.fit(X, y)

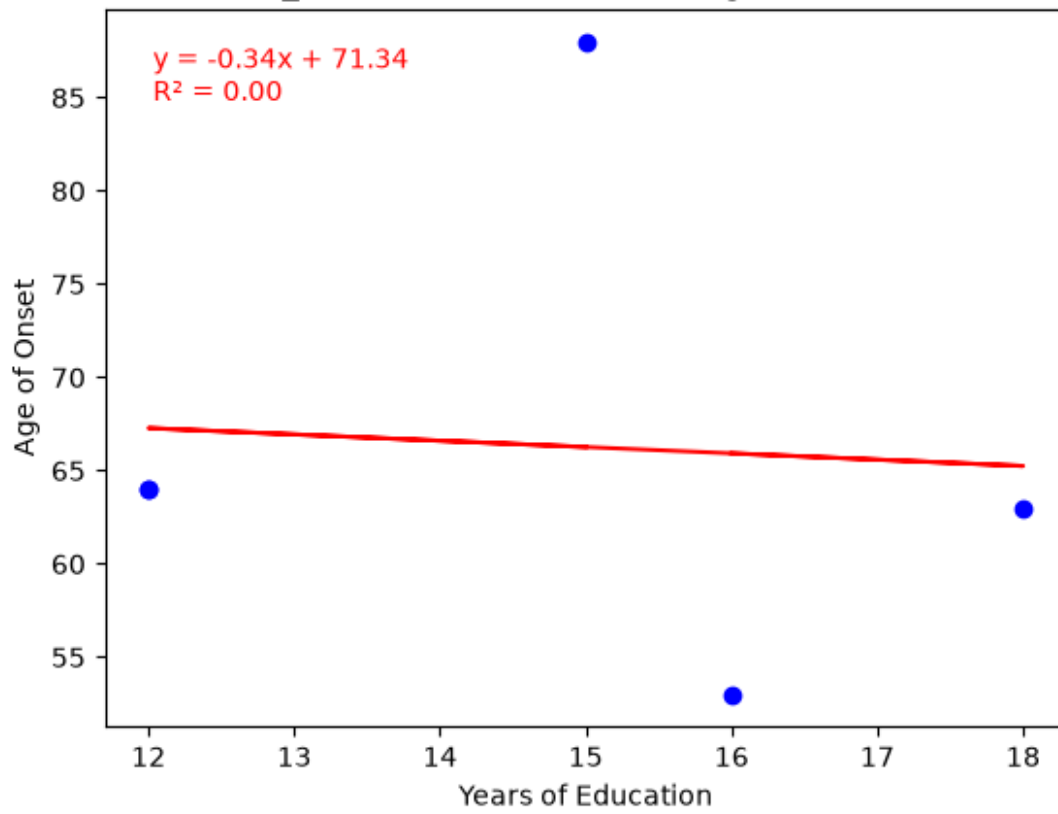
    slope = model.coef_[0]
    intercept = model.intercept_
    r2 = model.score(X, y)

    equation = f"y = {slope:.2f}x + {intercept:.2f}\nR² = {r2:.2f}"

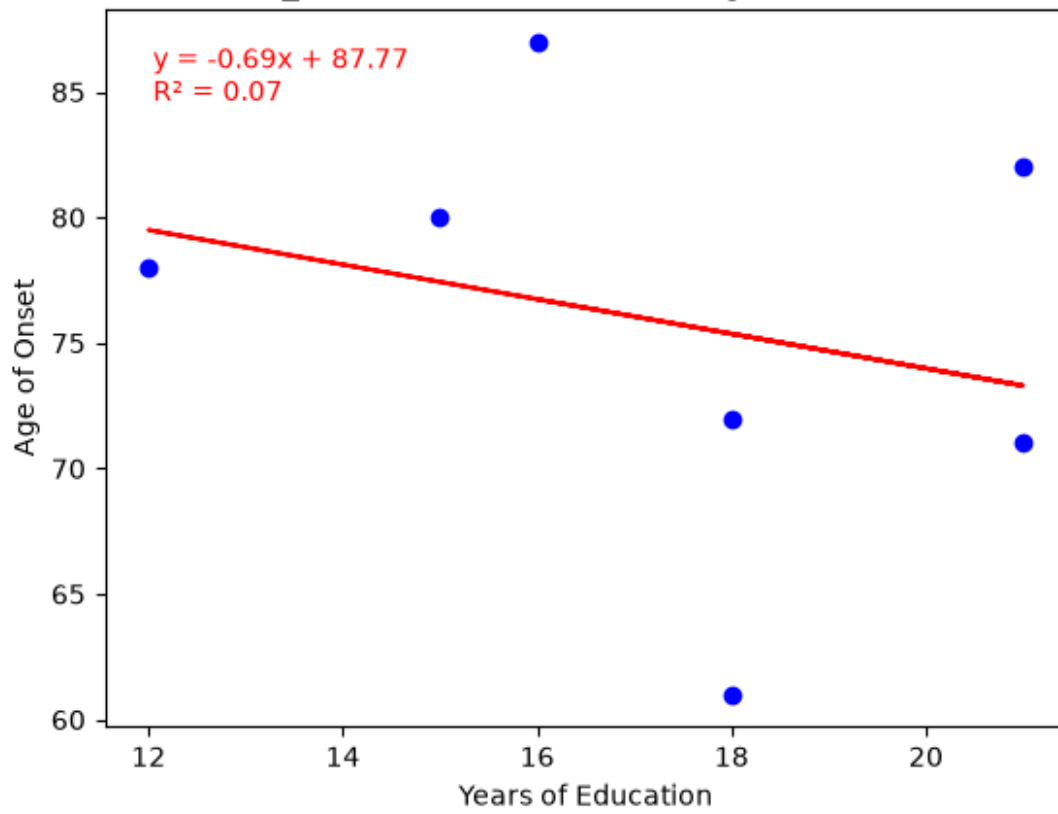
    plt.scatter(X[:, 0], y, color="blue")
    plt.plot(X[:, 0], model.predict(X), color="red")
    plt.text(
        0.05,
        0.95,
        equation,
        transform=plt.gca().transAxes,
        color="red",
        verticalalignment="top",
        bbox={"facecolor": "white", "alpha": 0.8, "edgecolor":
"none"},
    )
    plt.xlabel("Years of Education")
    plt.ylabel("Age of Onset")
    plt.title("3_2: Years of Education vs Age of Onset")
    plt.show()

```

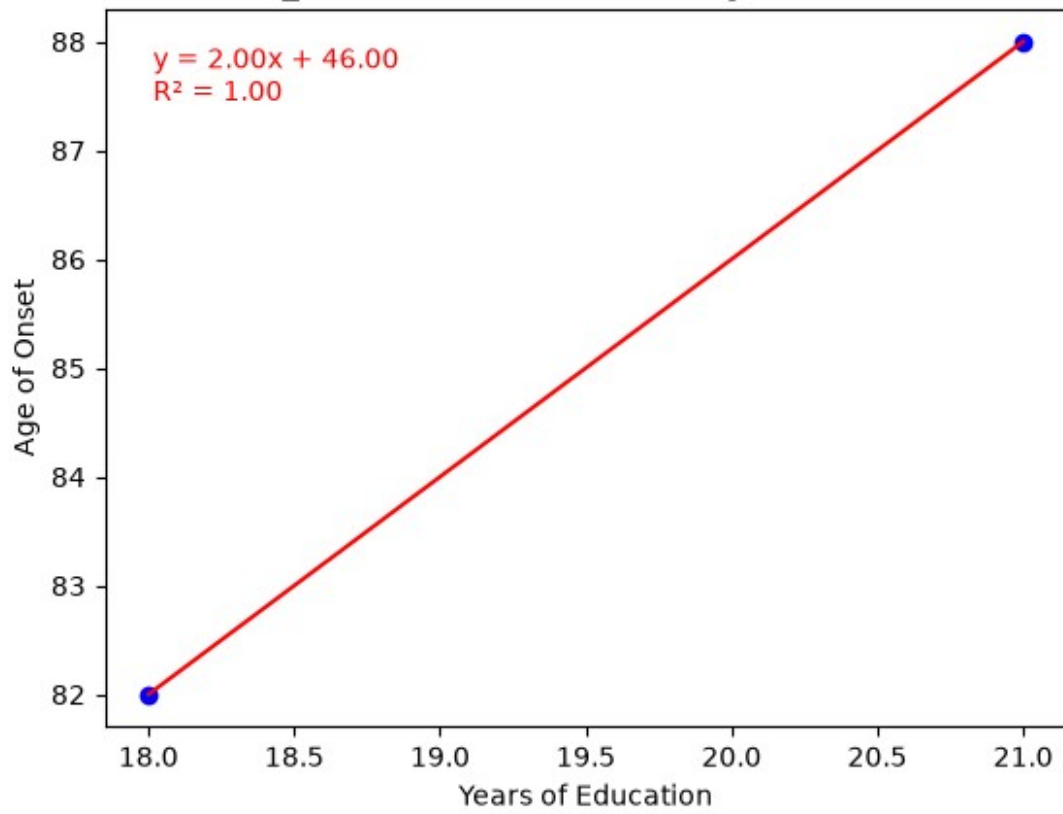
4\_4: Years of Education vs Age of Onset

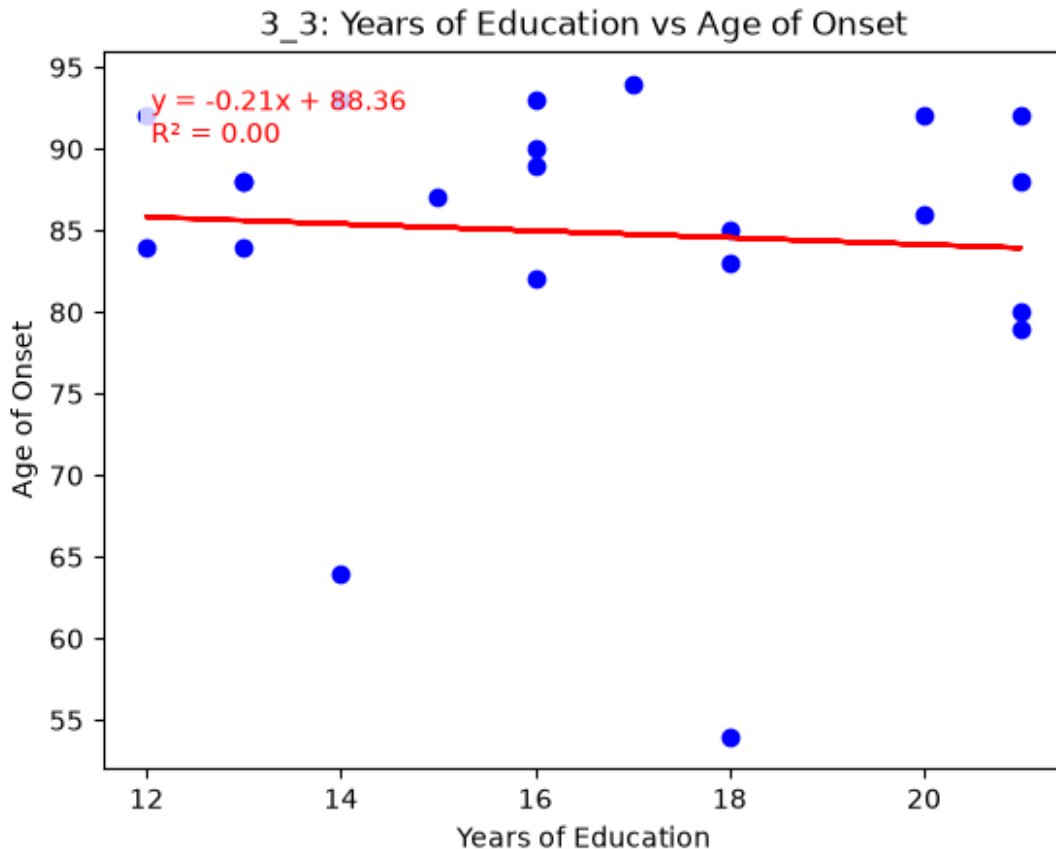


3\_4: Years of Education vs Age of Onset



2\_4: Years of Education vs Age of Onset





6. Create the empty list to store the data points of patients that have been educated within the required range of years

```
#-----
#Code that groups the patients by length of time they were educated
#First create the empty lists to store the data needed for the bar
graph
patients_0_13_years_educated_age_onset = []
patients_13_16_years_educated_age_onset = []
patients_16_20_years_educated_age_onset = []
patients_20_plus_years_educated_age_onset = []
```

7. Call methods created in Mod1\_main.py that sort through CSV file and filter the patients with the requested range of years of education. The groups of education are 0-13 years, 13-16 years, 16-20 years, 20+

```
#call the method made in the patient class that gets the number of
patients with 0-13 years of education
print("\nPatients with 0-13 years of education:\n")
for patient in Patient.get_patients_number_of_years_educated(0, 13):
```

```

    if patient.age_of_onset is not None:
        #this condition makes sure that only patients with a recorded
age of onset is added to the sub data set for 0-13 years of education

patients_0_13_years_educated_age_onset.append(patient.age_of_onset)
    print(patient)

#this block of code is the same as 252-257 but for 13-16 years of
education
print("\nPatients with 13-16 years of education:\n")
for patient in Patient.get_patients_number_of_years_educated(13, 16):
    if patient.age_of_onset is not None:

patients_13_16_years_educated_age_onset.append(patient.age_of_onset)
    print(patient)

#this block of code is the same as 252-257 but for 16-20 years of
education
print("\nPatients with 16-20 years of education:\n")
for patient in Patient.get_patients_number_of_years_educated(16, 20):
    if patient.age_of_onset is not None:

patients_16_20_years_educated_age_onset.append(patient.age_of_onset)
    print(patient)

#this block of code is the same as 252-257 but for 20+ years of
education
print("\nPatients with 20+ years of education:\n")
for patient in Patient.get_patients_number_of_years_educated(20, 100):
    if patient.age_of_onset is not None:

patients_20_plus_years_educated_age_onset.append(patient.age_of_onset)
    print(patient)

```

Patients with 0-13 years of education:

```

Gender: Female, Age of Onset: None, Gene Type: 2_3
Gender: Female, Age of Onset: None, Gene Type: 2_3
Gender: Male, Age of Onset: 64, Gene Type: 4_4
Gender: Male, Age of Onset: None, Gene Type: 3_3
Gender: Female, Age of Onset: 64, Gene Type: 4_4
Gender: Female, Age of Onset: 92, Gene Type: 3_3
Gender: Female, Age of Onset: 84, Gene Type: 3_3
Gender: Male, Age of Onset: None, Gene Type: 3_3
Gender: Female, Age of Onset: None, Gene Type: 3_3
Gender: Male, Age of Onset: None, Gene Type: 3_3
Gender: Female, Age of Onset: None, Gene Type: 2_3
Gender: Female, Age of Onset: 78, Gene Type: 3_4

```

Patients with 13-16 years of education:

Gender: Male, Age of Onset: 80, Gene Type: 3\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: 84, Gene Type: 2\_3  
Gender: Male, Age of Onset: 64, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: 88, Gene Type: 3\_3  
Gender: Female, Age of Onset: 94, Gene Type: 2\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: 88, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 2\_3  
Gender: Male, Age of Onset: None, Gene Type: 2\_3  
Gender: Male, Age of Onset: 93, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: 84, Gene Type: 3\_3  
Gender: Female, Age of Onset: 88, Gene Type: 4\_4  
Gender: Female, Age of Onset: 87, Gene Type: 3\_3

Patients with 16-20 years of education:

Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: 87, Gene Type: 3\_4  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: 83, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: 54, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 2\_2  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: 85, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 2\_3  
Gender: Female, Age of Onset: 63, Gene Type: 4\_4  
Gender: Female, Age of Onset: 61, Gene Type: 3\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 2\_3  
Gender: Male, Age of Onset: 89, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: 82, Gene Type: 3\_3  
Gender: Female, Age of Onset: 53, Gene Type: 4\_4

Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: 90, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: 82, Gene Type: 2\_4  
Gender: Male, Age of Onset: None, Gene Type: 3\_4  
Gender: Male, Age of Onset: 72, Gene Type: 3\_4  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: None, Gene Type: 2\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_4  
Gender: Female, Age of Onset: 93, Gene Type: 3\_3  
Gender: Male, Age of Onset: 94, Gene Type: 3\_3

Patients with 20+ years of education:

Gender: Female, Age of Onset: 71, Gene Type: 3\_4  
Gender: Female, Age of Onset: 92, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 4\_4  
Gender: Male, Age of Onset: 88, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 2\_3  
Gender: Female, Age of Onset: 80, Gene Type: 3\_3  
Gender: Female, Age of Onset: 88, Gene Type: 2\_4  
Gender: Female, Age of Onset: 82, Gene Type: 3\_4  
Gender: Female, Age of Onset: 92, Gene Type: 3\_3  
Gender: Female, Age of Onset: 86, Gene Type: 3\_3  
Gender: Male, Age of Onset: 79, Gene Type: 3\_3  
Gender: Female, Age of Onset: None, Gene Type: 3\_3  
Gender: Male, Age of Onset: None, Gene Type: 3\_3

## 8. Reformat list into arrays to fit the proper format of the statistical models

```
#Following the built in AIs Advice reformatting the list into arrays  
for the proper format for the statistics models  
x_patients_0_13_years_educated_age_onset =  
np.array(patients_0_13_years_educated_age_onset)  
x_patients_13_16_years_educated_age_onset =  
np.array(patients_13_16_years_educated_age_onset)  
x_patients_16_20_years_educated_age_onset =  
np.array(patients_16_20_years_educated_age_onset)  
x_patients_20_plus_years_educated_age_onset =  
np.array(patients_20_plus_years_educated_age_onset)
```

## 9. Find the mean age of onset in each group

```
#calculating the mean of each group using the np model for mean
patients_0_13_years_educated_age_onset_mean =
np.mean(x_patients_0_13_years_educated_age_onset)
patients_13_16_years_educated_age_onset_mean =
np.mean(x_patients_13_16_years_educated_age_onset)
patients_16_20_years_educated_age_onset_mean =
np.mean(x_patients_16_20_years_educated_age_onset)
patients_20_plus_years_educated_age_onset_mean =
np.mean(x_patients_20_plus_years_educated_age_onset)
```

## 10. Finding the standard deviation of each group

```
#calculating the std of each group using the std model
patients_0_13_years_educated_age_onset_std =
np.std(x_patients_0_13_years_educated_age_onset)
patients_13_16_years_educated_age_onset_std =
np.std(x_patients_13_16_years_educated_age_onset)
patients_16_20_years_educated_age_onset_std =
np.std(x_patients_16_20_years_educated_age_onset)
patients_20_plus_years_educated_age_onset_std =
np.std(x_patients_20_plus_years_educated_age_onset)
```

## 11. Print each group and the mean +/- the standard deviation for that respective group

```
#printing the results of the mean and standard deviation for each data group
print(f'\nAge of onset for 0-13 years of education:
{patients_0_13_years_educated_age_onset_mean} ±
{patients_0_13_years_educated_age_onset_std}')
print(f'\nAge of onset for 13-16 years of education:
{patients_13_16_years_educated_age_onset_mean} ±
{patients_13_16_years_educated_age_onset_std}')
print(f'\nAge of onset for 16-20 years of education:
{patients_16_20_years_educated_age_onset_mean} ±
{patients_16_20_years_educated_age_onset_std}')
print(f'\nAge of onset for 20+ years of education:
{patients_20_plus_years_educated_age_onset_mean} ±
{patients_20_plus_years_educated_age_onset_std}')
```

Age of onset for 0-13 years of education: 76.4 ± 11.056219968868202

Age of onset for 13-16 years of education: 85.0 ± 8.024961059095553

Age of onset for 16-20 years of education: 77.71428571428571 ± 13.843144623286447

Age of onset for 20+ years of education: 84.2222222222223 ± 6.4769296185731875

## 12. Organizing the values found in the previous block into the proper format for the bar graph model to run

```
#Creating the column names for the bar chart
year_educating = ['0-13', '13-16', '16-20', '20+']
#creating the arrays for the bar chart
mean_age_onset = [
    patients_0_13_years_educated_age_onset_mean,
    patients_13_16_years_educated_age_onset_mean,
    patients_16_20_years_educated_age_onset_mean,
    patients_20_plus_years_educated_age_onset_mean,
]
#creating the arrays of standard deviations for the bar chart
stdev_age_onset = [patients_0_13_years_educated_age_onset_std,
    patients_13_16_years_educated_age_onset_std,
    patients_16_20_years_educated_age_onset_std,
    patients_20_plus_years_educated_age_onset_std]

#This line is copied from the class example. It creates an asymmetric error bar which is the standard deviation bar that extends above each group in the bar graph
yerr = [np.zeros(len(mean_age_onset)), stdev_age_onset]
```

## 13. Format and display the bar graph

```
#This block of code formats the bar graph
plt.bar(year_educating, mean_age_onset, yerr=yerr, capsize=10,
    color=["blue", "orange"])
plt.title("Average Age of Onset Mediated by Years of Education")
plt.xlabel("Years of Education")
plt.ylabel("Average Age of Onset (age)")

#This block of code performs the one-way ANOVA test to determine if there are significant differences between the groups that would indicate level of education plays a role in deferring symptoms
f_stat, p_value =
stats.f_oneway(x_patients_0_13_years_educated_age_onset,
    x_patients_13_16_years_educated_age_onset,
    x_patients_16_20_years_educated_age_onset,
    x_patients_20_plus_years_educated_age_onset)
print("\nF-statistic:", f_stat)
print("\np-value:", p_value)
plt.text(
```

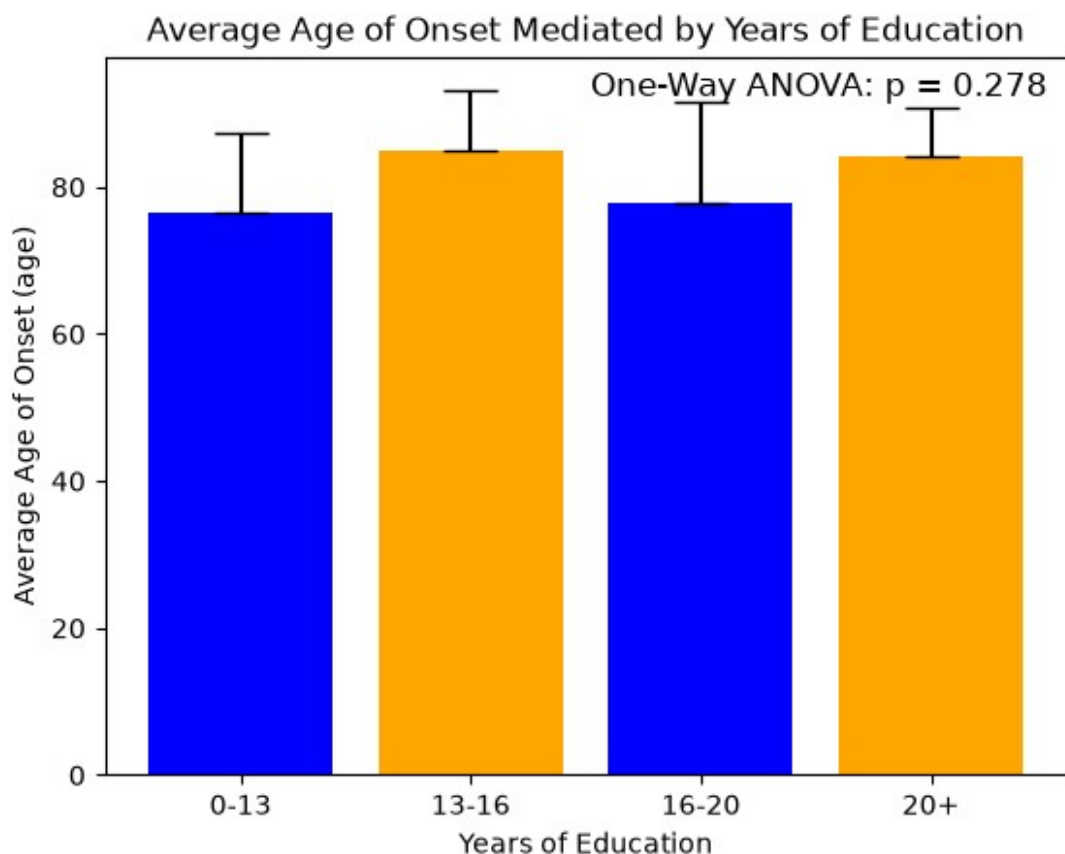
```

0.98,
0.98,
f"One-Way ANOVA: p = {p_value:.3f}",
transform=plt.gca().transAxes,
ha="right",
va="top",
fontsize=12,
)
plt.show()

```

F-statistic: 1.3403402534515998

p-value: 0.2775339842108476



## Verify and validate your analysis:

Verify: Prior to running our analysis, we hypothesized that the more years an individual was educated, the older they would be when they started showing signs of AD. We knew the sample sizes were small, but we still believed we would see some relationship. Therefore, the results are not what we expected. We think there is more nuance to the relationship between length of education and age of onset than this analysis describes. After reviewing other published

findings, we believe there is still a relationship between cognitive stimuli and symptoms of AD; they may just need to be measured and compared using alternative methods.

Validate: Published work gives some context, but none of it directly confirms our result. Sohrabi et al. (2025) also found no significant link between education and cognition in adults 80 and older, which is similar to our null result since our donors averaged about 88, although they measured cognition instead of age of onset. Baumgaertel et al. (2016) explain that for the cognitive reserve idea, education may not change the age of onset. APOE  $\epsilon$ 4 is an established risk factor for Alzheimer's (Corder et al., 1993), so we expected  $\epsilon$ 4 groups to show earlier onset, but our groups were too small to test that.

### Sources:

Baumgaertel, J., Haussmann, R., Gruschwitz, A., Werner, A., Osterrath, A., Lange, J., Donix, K. L., Linn, J., & Donix, M. (2016). Education and genetic risk modulate hippocampal structure in Alzheimer's disease. *Aging and Disease*, 7(5), 553-560. <https://doi.org/10.14336/AD.2016.0305>

Corder, E. H., Saunders, A. M., Strittmatter, W. J., Schmechel, D. E., Gaskell, P. C., Small, G. W., Roses, A. D., Haines, J. L., & Pericak-Vance, M. A. (1993). Gene dose of apolipoprotein E type 4 allele and the risk of Alzheimer's disease in late onset families. *Science*, 261(5123), 921-923. <https://doi.org/10.1126/science.8346443>

Sohrabi, H. R., Goozee, K. G., Weinborn, M., Rainey Smith, S., Taddei, K., & Martins, R. N. (2025). Does education years delay Alzheimer's disease in cognitively normal oldest old? *Alzheimer's & Dementia*. <https://doi.org/10.1002/alz.089080>

## Conclusions and Ethical Implications:

The first research question investigated whether the number of years an individual was educated had an impact on the age of onset of symptoms. We analyzed the data using an ANOVA test, which resulted in a p-value = 0.278, which is not less than the commonly used significance value of 0.05. Therefore, the ANOVA test failed and we failed to reject the null hypothesis. There is no significant difference between the means of the different groups, and therefore, the age of onset symptoms is not impacted by the number of years of education.

The second research question focused on the impact of years of education on the age of onset symptoms for each gene variant. The analysis method included graphing a scatterplot and applying a linear regression. Gene variants 4\_4, 3\_4, 3\_3, and 2\_4 were successfully graphed individually. None of the linear regressions resulted in a value of R<sup>2</sup> close to one, which means there is no relationship between the number of years of education and the age of onset of symptoms for patients with any gene variant.

There are limitations to our data and analysis and, therefore, inherently to our conclusions. When examining the different gene variants, we were not able to analyze gene variants 3\_2 and 2\_2 because there were not enough patients in the data set to properly run a linear regression. Furthermore, across the other genes, the number of data points was way too small to draw an

accurate conclusion. Therefore, the conclusions drawn between the number of years of education and age of onset within specific gene variants have very low validity.

The data analysis for the first research question has a considerable number of data points. While increasing the number of data points added would increase the validity of the conclusion, the conclusion we drew is still worth examining. If there is no significant impact of years of education on the age of onset of symptoms, it can be argued that continual cognitive stimuli does not defer the symptoms of Alzheimer's. In this case, medical professionals and researchers should focus their attention on other possible mediators that could delay symptoms. The ethical implication is the possible consequences and successes resulting from redirecting researchers' focus to treatments that have a higher chance of managing symptoms, which would require reducing the amount of research on the role of cognitive stimuli.

## Limitations and Future Work:

The biggest limitation in this data set, and therefore our analysis, is the small number of data points. As mentioned in the conclusion section, the lack of data for the individual gene variants nullifies the conclusions drawn from the linear regression model.

For the data analysis of the entire data set, comparing the number of years of education and age of onset of symptoms, an increase in data points would improve the strength of the conclusion. Additionally, the number of years of education is incredibly vague. There is no way to know the type of education, the difficulty level, and therefore the level of cognitive stimuli the education provided. Without these details, it is very hard to generalize the findings to all forms of cognitive stimuli. In other words, it would not be surprising if there were studies that contradicted these findings because they had a more accurate way to measure cognitive stimuli.

In the future, to further analyze the relationship between the number of years of education and the age of onset of symptoms, we would start by using a much bigger data set. We would then be very specific about what was considered cognitive stimuli. We would also adjust for cultural differences. We would need to know what each culture considered cognitive stimuli to accurately compare patient data between cultures. For example, if we were to find data from a study similar to Asano, Taki, Hashizume et al. 2014, we would have to be careful about analyzing the amount of studying each student does because there could be discrepancies about how intense studying should be across cultures.

## AI Usage Statement:

The only AI used to create the code was the built in AI in VsCode. The AI was used to debug the code, and help with formatting the bar graphs and scatter plots. Gemini, Claude and Google AI were used to assist with research. There are separate AI Usage Statements in the background section that elaborate

## GitHub Repository Link:

([https://github.com/JaneW-14/BME2315\\_Module1\\_Project](https://github.com/JaneW-14/BME2315_Module1_Project))